

ClipMind AI: Video Summarization & KeyMoments Detection

Milestone 2: Key Moments Detection, Analytics, Testing and Deployment (Module 3 and Module 4)

1. My Work Responsibility

- **Module 3 (Key Moments Detection and Analytics):** Backend APIs for Key Moments, Highlights, Keywords and Analytics; API and schema validation; response-field consistency; error handling; authentication verification; API testing.
- **Module 4 (Testing, Deployment and Documentation):** API performance testing, video processing optimization support, Docker and Docker Compose setup, and backend deployment.

2. Project Objective

The objective of Milestone 2 is to extend ClipMind AI from transcription and summarization to full video intelligence. Module 3 analyzes the transcript to identify the most important portions of a video, with their timestamps and topics, and exposes the results together with keywords and analytics. Module 4 validates the AI outputs, tests and optimizes performance, and prepares the platform for deployment, documentation and the final demonstration.

My objective in this milestone was to build and verify the backend API layer that provides key moments, highlights, keywords and analytics, and to prepare the backend for containerized deployment.

3. Project Description and Main Features

In Milestone 2, the platform works on the transcript generated in the previous milestone. The complete processing flow is:

Upload, Processing, Transcript with timestamps, AI Summary, Key Moments, Highlights, Keywords, Analytics, Display of results.

Module 3 Requirements: Key Moments Detection

Module 3 takes the timestamped transcript from Module 2 and identifies the important parts of the video. The requirements are:

- 1 **Transcript segmentation:** divide the transcript into smaller segments that keep their start and end timestamps, respecting sentence boundaries.
- 2 **Embeddings:** convert segments into numerical vectors using a pretrained embedding model (Sentence Transformers).
- 3 **Similarity:** compare neighboring segments with cosine similarity to see whether they discuss related content.
- 4 **Topic segmentation:** use similarity changes between neighboring segments to find where one topic ends and the next begins.
- 5 **Importance scoring:** score each segment using factors such as semantic relevance, keywords, topic relevance, content density and position.

- 6 **Highlight extraction:** rank segments, select the most useful ones, and avoid duplicate highlights from overlapping segments.
- 7 **Stored information:** for each key moment, store the video ID, topic, start time, end time, transcript text and importance score.

Module 3 Features: Backend APIs

I built the backend APIs that provide the Module 3 results:

- **Key Moments API:** accepts the processed video and transcript data and returns the detected key moments with timestamps and importance scores.
- **Highlights API:** returns the top-ranked highlights for a video.
- **Keywords API:** returns the important keywords and topics extracted from a video's transcript.
- **Analytics API:** returns the analytics of a video, including its processing status, transcript, short summary and summary word counts, key moment, highlight and keyword counts, the top keywords, and the upload time.
- **Consistent responses:** the APIs return consistent JSON with stable field names across all endpoints.
- **Authentication:** all endpoints are protected with JWT, and the login flow was re-verified end to end, including OAuth2 form-data handling.
- **Error handling:** bad uploads, unsupported formats, empty transcripts and missing data return clear error responses.
- **Database integration:** the APIs read and write the key-moment, keyword and analytics data in MongoDB, linked to the correct video.

Module 4 Requirements: Testing, Deployment and Documentation

Module 4 prepares the platform for final submission. The requirements are:

- 1 Validate transcript accuracy and summary quality.
- 2 Optimize the video processing pipelines.
- 3 Deploy the platform using Docker and cloud environments.
- 4 Prepare the final project documentation and presentation.
- 5 Demonstrate the complete ClipMind AI platform end to end.

Module 4 Features: Testing and Deployment

My part in Module 4 covered the backend side of testing and deployment:

- **API testing:** every endpoint was tested with Postman for valid input, invalid input, missing data and authentication failures.
- **API performance testing:** response times of the APIs were measured and checked.
- **Processing optimization support:** processing time from upload to results was measured to support optimization of the pipeline.
- **Docker setup:** a Docker Compose file runs three services: MongoDB, the FastAPI backend (built from the backend folder with its environment file), and a Caddy web server on ports 80 and 443.

Named volumes keep the MongoDB data, uploaded videos, generated clips, and the Hugging Face and Whisper model caches. This is the base for backend and cloud deployment.

- **API documentation:** endpoints and their request and response shapes were documented for the final documentation package.

4. Technology Used

- **Language:** Python
- **Backend Framework:** FastAPI with Uvicorn
- **Database:** MongoDB Atlas with Motor and Beanie
- **Authentication:** JWT, with OAuth2 password form handling for login
- **Validation and Schemas:** Pydantic
- **API Testing:** Postman and Swagger UI
- **Containerization:** Docker and Docker Compose (MongoDB, backend and Caddy services)
- **Version Control:** Git and GitHub, with feature branches and Pull Requests
- **Development Tool:** Visual Studio Code

5. Screenshots

Figure 1: Key Moments API response (timestamps, topics and importance scores)

http://127.0.0.1:8000/api/videos/6ac8900c92590fae0bb05a71/key-moments

Server response

Code	Details
200	Response body <pre>{ "id": "6ac8900c92590fae0bb05a71", "filename": "test_video2.mp4", "status": "done", "current_stage": "done", "progress": 100, "upload_progress": 100, "audio_progress": 100, "transcription_progress": 100, "summary_progress": 100, "key_moments_progress": 100, "highlights_progress": 100, "keywords_progress": 100, "transcription_language": "en", "transcript": "What is machine learning and how does it work? Netflix recommends shows you love. Instagram knows which reels might keep you scrolling. But how do they know? That's machine learning. It's one of the most exciting parts of data science and it's reshaping industries like healthcare, finance, and transportation. Instead of writing every rule for a computer, we show it examples. It learns patterns from data, show it thousands of emails, marked spam or not spam, and it learns the difference. Then when a new email arrives, it predicts spam or not spam. It's the same trick behind recommendations, fraud alerts, and personalized apps. Machine learning finds patterns in data and makes smarter predictions. In the next video, I'll explain AI, machine learning, and deep learning.", "transcript_segments": [{ "start_time": 0, "end_time": 2.2, "text": "What is machine learning and how does it work?" }, { "start_time": 2.2,</pre> Response headers <pre>"key_moments": [{ "start_time": 0, "end_time": 45.8, "label": "What is machine learning and how does it...", "text": "What is machine learning and how does it work? Netflix recommends shows you love. Instagram knows which reels might keep you scrolling. But how do they know? That's machine learning. It's one of the most exciting parts of data science and it's reshaping industries like healthcare, finance, and transportation. Instead of writing every rule for a computer, we show it examples. It learns patterns from", "importance": 0.834 }],</pre>

Figure 2: Highlights and Keywords API responses

Request URL	
http://127.0.0.1:8000/api/videos/6ac8900c92590fae0bb05a71/highlights	
Server response	
Code	Details
200	Response body <pre>{ "id": "6ac8900c92590fae0bb05a71", "filename": "test_video2.mp4", "status": "done", "current_stage": "done", "progress": 100, "upload_progress": 100, "audio_progress": 100, "transcription_progress": 100, "summary_progress": 100, "key_moments_progress": 100, "highlights_progress": 100, "keywords_progress": 100, "transcription_language": "en", "transcript": "What is machine learning and how does it work? Netflix recommends shows you love. Instagram knows which reels might keep you scrolling. But how do they know? That's machine learning. It's one of the most exciting parts of data science and it's reshaping industries like healthcare, finance, and transportation. Instead of writing every rule for a computer, we show it examples. It learns patterns from data, show it thousands of emails, marked spam or not spam, and it learns the difference. Then when a new email arrives, it predicts spam or not spam. It's the same trick behind recommendations, fraud alerts, and personalized apps. Machine learning finds patterns in data and makes smarter predictions. In the next video, I'll explain AI, machine learning, and deep learning.", "transcript_segments": [{ "start_time": 0, "end_time": 2.2, "text": "What is machine learning and how does it work?" }, { "start_time": 2.2,</pre>

```

    },
    {
      "start_time": 2.2,
      "end_time": 45.8,
      "text": "What is machine learning and how does it work? Netflix recommends shows you love. Instagram knows which reels might keep you scrolling. But how do they know? That's machine learning. It's one of the most exciting parts of data science and it's reshaping industries like healthcare, finance, and transportation. Instead of writing every rule for a computer, we show it examples. It learns patterns from data, show it thousands of emails, marked spam or not spam, and it learns the difference. Then when a new email arrives, it predicts spam or not spam. It's the same trick behind recommendations, fraud alerts, and personalized apps. Machine learning finds patterns in data and makes smarter predictions. In the next video, I'll explain AI, machine learning, and deep learning.",
      "importance": 0.834
    }
  ],
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      "importance": 0.834
    }
  ],
  "keywords": [
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      "word": "machine",
      "count": 126,
      "rank": 1
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    {
      "word": "learning",
      "count": 122,
      "rank": 2
    },
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      "count": 44,
      "rank": 3
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      "count": 10,
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      "count": 10,
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      "rank": 10
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      "count": 10,
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      "rank": 48
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      "count": 10,
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      "count": 10,
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    {
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      "rank": 57
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      "count": 10,
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      "count": 10,
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      "count": 10,
      "rank": 70
    },
    {
      "word": "deep",
      "count": 10,
      "rank": 71
    },
    {
      "word": "learning",
      "count": 10,
      "rank": 72
    }
  ],
  "uploaded_at": "2026-10-09T06:56:12.113000"
}
```

Figure 3: Analytics API response (word counts, key moment, highlight and keyword counts, and top keywords for a video)

Request URL	
http://127.0.0.1:8000/api/analytics/videos/6ac8900c92590fae0bb05a71	
Server response	
Code	Details
200	Response body <pre>{ "video_id": "6ac8900c92590fae0bb05a71", "filename": "test_video2.mp4", "status": "done", "transcript_word_count": 126, "short_summary_word_count": 44, "summary_word_count": 122, "key_moments_count": 1, "highlights_count": 1, "keywords_count": 10, "top_keywords": [{ "word": "machine", "count": 126, "rank": 1 }, { "word": "learning", "count": 122, "rank": 2 }, { "word": "data", "count": 44, "rank": 3 }, { "word": "it", "count": 10, "rank": 4 }, { "word": "the", "count": 10, "rank": 5 }, { "word": "and", "count": 10, "rank": 6 }, { "word": "how", "count": 10, "rank": 7 }, { "word": "does", "count": 10, "rank": 8 }, { "word": "work", "count": 10, "rank": 9 }, { "word": "that", "count": 10, "rank": 10 }, { "word": "is", "count": 10, "rank": 11 }, { "word": "a", "count": 10, "rank": 12 }, { "word": "computer", "count": 10, "rank": 13 }, { "word": "we", "count": 10, "rank": 14 }, { "word": "show", "count": 10, "rank": 15 }, { "word": "it", "count": 10, "rank": 16 }, { "word": "examples", "count": 10, "rank": 17 }, { "word": "it", "count": 10, "rank": 18 }, { "word": "learns", "count": 10, "rank": 19 }, { "word": "patterns", "count": 10, "rank": 20 }, { "word": "from", "count": 10, "rank": 21 }, { "word": "data", "count": 10, "rank": 22 }, { "word": "show", "count": 10, "rank": 23 }, { "word": "it", "count": 10, "rank": 24 }, { "word": "thousands", "count": 10, "rank": 25 }, { "word": "of", "count": 10, "rank": 26 }, { "word": "emails", "count": 10, "rank": 27 }, { "word": "marked", "count": 10, "rank": 28 }, { "word": "spam", "count": 10, "rank": 29 }, { "word": "or", "count": 10, "rank": 30 }, { "word": "not", "count": 10, "rank": 31 }, { "word": "spam", "count": 10, "rank": 32 }, { "word": "and", "count": 10, "rank": 33 }, { "word": "it", "count": 10, "rank": 34 }, { "word": "predicts", "count": 10, "rank": 35 }, { "word": "spam", "count": 10, "rank": 36 }, { "word": "or", "count": 10, "rank": 37 }, { "word": "not", "count": 10, "rank": 38 }, { "word": "spam", "count": 10, "rank": 39 }, { "word": "It's", "count": 10, "rank": 40 }, { "word": "the", "count": 10, "rank": 41 }, { "word": "same", "count": 10, "rank": 42 }, { "word": "trick", "count": 10, "rank": 43 }, { "word": "behind", "count": 10, "rank": 44 }, { "word": "recommendations", "count": 10, "rank": 45 }, { "word": "fraud", "count": 10, "rank": 46 }, { "word": "alerts", "count": 10, "rank": 47 }, { "word": "and", "count": 10, "rank": 48 }, { "word": "personalized", "count": 10, "rank": 49 }, { "word": "apps", "count": 10, "rank": 50 }, { "word": "Machine", "count": 10, "rank": 51 }, { "word": "learning", "count": 10, "rank": 52 }, { "word": "finds", "count": 10, "rank": 53 }, { "word": "patterns", "count": 10, "rank": 54 }, { "word": "in", "count": 10, "rank": 55 }, { "word": "data", "count": 10, "rank": 56 }, { "word": "and", "count": 10, "rank": 57 }, { "word": "makes", "count": 10, "rank": 58 }, { "word": "smarter", "count": 10, "rank": 59 }, { "word": "predictions", "count": 10, "rank": 60 }, { "word": "In", "count": 10, "rank": 61 }, { "word": "the", "count": 10, "rank": 62 }, { "word": "next", "count": 10, "rank": 63 }, { "word": "video", "count": 10, "rank": 64 }, { "word": "I'll", "count": 10, "rank": 65 }, { "word": "explain", "count": 10, "rank": 66 }, { "word": "AI", "count": 10, "rank": 67 }, { "word": "machine", "count": 10, "rank": 68 }, { "word": "learning", "count": 10, "rank": 69 }, { "word": "and", "count": 10, "rank": 70 }, { "word": "deep", "count": 10, "rank": 71 }, { "word": "learning", "count": 10, "rank": 72 }], "uploaded_at": "2026-10-09T06:56:12.113000" }</pre>

6. Challenges Faced

- **Dependency on upstream output.** The Key Moments API could not be finalized until the timestamp extraction and key-moment logic produced real output. I designed the API structure first and connected it to the real output as it became stable.
- **Inconsistent field names.** Field names in the API schemas and responses were not always consistent. I validated the schemas against real data and made the response fields consistent.
- **Login form-data handling.** The login endpoint needed to accept OAuth2 form-data and return an access token. I fixed the handling and re-verified the full authentication flow.

- **Analytics numbers not matching the data.** Aggregate counts had to agree with the underlying video, key-moment and keyword records. I verified the computed values against manually counted results and fixed the inconsistencies.
- **Short time for integration.** Several members' work had to be combined within a short window. Daily merges through Pull Requests kept the code in a working state.

7. Conclusion

This milestone extended ClipMind AI with key moments detection, highlights, keywords and analytics. On the backend, I delivered the Key Moments, Highlights, Keywords and Analytics APIs, secured with JWT, connected to MongoDB, and returning consistent responses with proper error handling. The APIs were tested with Postman to confirm correct behavior for valid input, invalid input and authentication failures.